Case Study 2

Data Cleaning for Student Performance Dataset

Date of the Session: ....

Learning outcomes:

- understand the descriptive statistics: Mean, Median, Mode;
- data cleaning - replacing the missing values with the mean, median, and mode
- Comparison of data metrics before and after cleaning

2.1 Pre-Lab

Define the terms of descriptive statistics.

1. Mean → Sum of all values ÷ no. of values.
2. Median → Avg (or) middle value.
3. Mode → Most frequently occurring term.
4. Missing values → Absence (or) unavailable of data.
5. outliers → Values that are different from set.

2.2 Problem Statement

This data approach to student achievement in secondary education of two Portuguese schools. The data attributes include student grades, demographic, social and school-related features and it was collected by using school reports and questionnaires.

Two datasets are provided regarding the performance in two distinct subjects: Mathematics (mat.csv) and Portuguese language (por.csv). In [Cortez and Silva, 2008], the two datasets were modeled under binary/five-level classification and regression tasks. The dataset is available in the link ☞student performance.

The following analyses are to be performed on the dataset:

1. Compute the mean of the travel time and replace the missing values of the travel time.

2. Compute the median of the travel time and replace the missing values of the travel time.

Sol Median (after Sorted) = 1

Median is 1 → replaced with missing values.

3. Analyze the mean of the travel time before replacement, replacement with mean and median. Comment

A) Replacement with mean : Missing values are filled with 1.474

New mean = 1.474 (unchanged)

Replacement with median : Missing values filled with 1

New mean = 1.475

Slightly decrease in mean because median is lower than mean.

Comments :

i) The original mean (1.474) indicates most students have low travel time.

ii) Median = 1.0 Confirms that more than half of student fall in the shortest travel category.

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Evaluators Comments	Evaluator's Observation
	Marks Secured .... out of 50
	Full Name of the Evaluator:
	Signature of the Evaluator:
	Date of Evaluation:

Case Study 3

E-Commerce Customer Analytics

Date of the Session: ....

Learning outcomes:

- Identification of type of variable in the dataset
- Understanding the dataset
- Computation of feature metrics of the dataset

3.1 Pre-Lab

Define the terms of descriptive statistics.

1. Range → highest value - lowest value.
2. Standard deviation → Square root of variance
3. Quartile → It divides data into 4 parts ($Q_1 \to 25\%$. $Q_2 \to 50\%$. $Q_3 \to 75$
4. Inter Quartile Range → IQR = $Q_3 - Q_1$ (Middle spread value).

3.2 Problem Statement

An online retailer collects the following data for 10 customers:

Table 3.1: Customer Data

Customer	Age	Gender	Monthly Spend	Orders	Satisfaction Rating (1–5)
C1	22	F	2500	5	4
C2	35	M	4000	8	5
C3	28	F	3200	6	3
C4	40	M	5000	10	5
C5	30	F	2800	5	4
C6	27	M	3500	7	3
C7	45	F	6000	12	5
C8	38	M	4200	9	4
C9	26	F	3000	6	2
C10	33	M	3700	7	4

1. Classify each variable by data type and measurement scale.

A) customer → object
age → int
gender → object
Monthly spend → int
order → int
Satisfaction rating → int

2. Compute mean, median, and mode of Monthly Spend.

A) Mean = 3790.0

Median = 3600.0

No mode Exist

3. Calculate range, variance, and standard deviation of Monthly Spend.

A) Range = 3500

Variance = 1141000.0

Standard deviation = 1068.

4. Determine Q1, Q2, Q3 and IQR.

Q1 = 3050.0
IQR = 1100.0

Q2 = 3600.0

Q3 = 4150.0

5. Identify outliers using IQR method.

Lower bound = 1400.0

Upper bound = 5800.0

6. Analyse whether the data is skewed or symmetric.

Column wise :

Age = 0.377993 → Approx → Symmetric

Monthly spend = 1.003168 → Right skewed

orders = 0.851499 → Right skewed

satisfaction Rating = -0.610141 → Left skewed.

Row wise :

0 → 1.999736
1 → 1.999725
2 → 1.999707
3 → 1.999769
4 → 1.999554
5 → 1.999783
6 → 1.999796
7 → 1.999692
8 → 1.999704
9 → 1.999701

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Case Study 4

Dataset Supermarket Pricing Decision

Date of the Session: ....

Learning outcomes:

- Understand data spread using range, variance, and standard deviation.
- Compute quartiles (Q1, Q2, Q3) and IQR
- Identify outliers using the IQR method

4.1 Pre-Lab

1. What is range? How does it differ from variance and standard deviation?

A) Range = Highest value - lowest value. The diff, is range use 2→values, while varience & std Dev all values.

2. Define variance and standard deviation. What do they indicate about data distribution?

A) Variance mean, avg of squared difference from mean.
std Dev means squareroot of variance

3. What are quartiles (Q1, Q2, Q3)? How is the IQR calculated?

A) quartiles divide data into 4 parts,

Q1 → 25%, Q2 → 50%, Q3 → 75%.

IQR = Q3 - Q1 → Measure spread of middle data.

4.2 Problem Statement

A supermarket chain in a metropolitan city is reviewing the pricing of a popular grocery item across its outlets. The prices (Rs.) recorded from 12 stores are: 120, 135, 150, 150, 165, 180, 200, 220, 250, 300, 320, 1200 The store manager notices that one outlet has priced the item significantly higher than others. Tasks:

1. Identify the type and measurement scale of the data collected.

A) Data type : Quantitative / Continous

Scale : Ratio (true zero Exist price Cannot be -ve)

2. Compute the mean, median, and mode of the prices.

A) Mean = 282.5

Median = 190.0

Mode = 150 → Repeated Couple of times.

3. The company wants to advertise a "typical price" to customers. Which measure should be used? Justify your choice based on the dataset.

A) Recommened measure : Median (190.0)
Reason : The Extream value is 1200 Inflats the mean to 282.5
Median is unaffected by outliers and represent more realistic & typical price

4. Calculate the range, variance, and standard deviation.

A) Range = 1080

Variance = 87534.09091

Standard deviation = 295.8616076970316

5. Determine Q1, Q2, Q3 and IQR.

A) $Q_1 = 150.0$

$Q_2 = (\text{Median}) = 190.0$

$Q_3 = 262.5$

$IQR = Q_3 - Q_1$

$= 112.5$

6. Identify any outliers using the IQR method.

A) Lower bound = -18.75

Upper bound = 431.25

7. Analyse how the extreme price influences business perception and suggest corrective action

A) Impact of Extream high prices

Mean with outlier = 282.5

Mean without outliers = 199.09

Corrective actions:

1. Investigate the store pricing Rs 1200 - likely data may be entered wrong

2. Audit pricing policies across all outlets.

3. If legitimate, separate it as premium product category.

Case Study 5

Student Performance and Academic Intervention

Date of the Session: ....

Learning outcomes:

- Classify type and scale of data in student performance
- Compute mean, median, standard deviation, and quartiles
- Identify cutoff scores for academic intervention and scholarships

5.1 Prelab

1. What are mean, median, and standard deviation, and why are they important?

A) Mean ⇒ Sum of all values ÷ no. of values.

Median ⇒ Middle value (Sorted order).

Std Dev is square root of varience.

2. What are quartiles and percentiles, and how are they calculated?

A) Quartiles divide data into 4 Equal parts, $Q_1 = n+1/4$

$n+1/2$, $3(n+1)/4$

$\downarrow$ Q_2 $\downarrow$ Q_3

Percentiles divide data into 100 Equal parts.

$$P_k = \frac{k(n+1)}{100}.$$

3. How can statistical measures help in identifying weak and high-performing students?

A) By Spearatiy weak & high performing students using marks data.

=> Mean, Median, std dev, Quartiles & percentiles.

5.2 Problem Statement

A university records marks (out of 100) for students in a course: 35, 40, 45, 50, 55, 60, 65, 70, 75, 80, 85, 90 The academic committee plans interventions for weak and high-performing students. Tasks:

1. Identify the type and scale of the data.

A) Datatype : Quantitative | Discreat

Scale : Ratio (0 marks is meaning full true zero)

2. Compute mean, median, and standard deviation.

A) mean = 62.5

Median = 62.5

standard devation = 18.027756377319946

3. Determine quartiles (Q1, Q2, Q3).

A) Q1 = 48.75

Q2 = 62.5

Q3 = 76.25

4. Students scoring below Q1 require remedial classes: Identify the cutoff score.

A) Remidial class cutoff (Q1) = 48.75

Students requiring remidial class:

	Student	Marks
0	S1	35
1	S2	40
2	S3	45

5. Students above the 75th percentile will receive scholarships: Determine the threshold.

A) Scholarship threshold ($Q_3/75^{th}$) = 76.35

Student eligible for scholarship are,

	Student	Marks
9	S10	80
10	S11	85
11	S12	90

6. Analyse the spread and consistency of marks.

A) IQR = 27.5

CV = 28.84%

7. Recommend strategies for improving overall performance

A) i) Introduce peer tutoring for students scoring below Q_1 = 48.75

ii) provide advanced coursework for scholarship holders (above Q_3 = 76.25)

iii) Track performance trends ($G_1 \to G_2 \to G_3$) for early intervention

iv) Conduct doubt clearing session for Middle (50%).

Case Study 6

Hospital Waiting Time Analysis

Date of the Session: ....

Learning outcomes:

- Understand different types of scales of measurement
- Identification of outliers and their effect on the mean.
- Replacement of outliers by mean

6.1 Pre-Lab

1. Define sample and identify its type.

A) Sample is a small part of population selected to represent whole. Types of samples are,
i) probability & Non-probability

2. What are outliers?

A) outliers are the data values which are different from rest of the values.

6.2 Problem Statement

A hospital records patient waiting times (in minutes) in the outpatient department: 10, 15, 20, 25, 30, 35, 40, 45, 50, 120 The hospital management claims that waiting times are reasonable. Answer the queries about patients' waiting times.

1. Identify the data type and measurement scale.

A) Data type: quantitative | Continous
scale: Ratio (0 min = no wait, true zero exists)

2. Mean and median waiting times.

A) Mean of waiting time = 29.0

Median of waiting time = 32.5

3. Evaluate whether the hospital's claim is justified.

A) Mean = 39.0 min (inflated by outliers 120 min)

Median = 32.5 min (true representative value)

Conclusion: Hospital claim is not fully satisfied. At least one patient waited 120 min extreme time.

4. Calculate range, standard deviation, and IQR.

A) Range = 110

Standard deviation = 31.2516666222459

$Q_1 = 21.25$ $Q_3 = 43.75$

$Q_2 = 23.5$

5. Identify outliers.

A) lower bound = -12.5

upper bound = 77.5

6. Analyse variability in service delivery.

A) CV = 80.13%.

Interpretation: High CV → Inconsistent service delivery.

7. Improvements to reduce waiting time inconsistencies.

A) i) investigate why a patient waiting 120 min.

ii) Introduce triage protocols to prioritize urgent cases.

iii) Add staff during peak hours to reduce stress.

Case Study 7

Manufacturing Process Quality Control

Date of the Session: ....

Learning outcomes:

- Identify and classify data types (discrete/continuous) and measurement scales used in real-world quality control scenarios.
- Apply descriptive statistical measures such as mean, median, and mode to summarize manufacturing defect data.
- Analyze variability using variance, standard deviation, quartiles, and interquartile range (IQR) to understand process dispersion.

7.1 Pre-Lab

1. What is the difference between discrete and continuous data? Give examples related to manufacturing.

A) discreate data means Countable data. Ex: whole number (7)

Continous data means measurable data. Ex: height (6.7)

2. Define mean, median, and mode. In what situations is each measure most useful?

A) Mean → sum of all values ÷ number of values.

Median → Middle value (data sorted).

Mode → Most frequently occuring value.

⇒ When data is normal (no Extream values) → Mean

⇒ When data has outliers & (or) skewness → Median

⇒ For most Common values (or) Categorical → Mode

3. Explain the concept of quartiles and interquartile range (IQR). Why are they important in statistical analysis?

A) Quartiles Split the data into 4 parts. (Q_1 Q_2 Q_3) 25% 50% 75%.

IQR: $Q_3 - Q_1$. They are important because, Detect outliers, show data spread clearly.

7.2 Problem Statement

A factory monitors the number of defective items per batch: 1, 2, 2, 3, 3, 3, 4, 4, 5, 6, 2, 3, 10 The quality control team wants to assess process stability. Tasks:

1. Identify whether the data is discrete or continuous and its scale.

A) Data type: Discrete (whole number)

Scale: (Ratio) (0 defects = perfect batch, true zero)

2. Compute mode, mean, and median.

A) Mode = [3]

Mean = 3.6923076923076925

Median = 3.0

3. Explain the significance of mode in this context.

A) Mode = 3. Most batches produce exactly 3 defects.

=> Mode reveals the most common defect level - critical for quality Control.

=> Production should target for mode ≤1. For better output.

4. Calculate variance and standard deviation.

Variance = 5.397435897435898

Standard deviation = 2.3232382351872 35

5. Determine quartiles and IQR.

A) $Q_1 = 2.0$

$Q_2 = 3.0$ (Median)

$Q_3 = 4.0$

IQR = 2.0 $(Q_3 - Q_1)$

6. Detect outliers.

A) lower bound = -1.0

upper bound = 7.0

7. whether the manufacturing process is stable and suggest improvements.

A) process stability is unstable

i) Investigate the batch producing 10 defects.

ii) Retrain operators on machinery causing frequent defects.

iii) Set Reject threshold : batches >8 should have be re-inspected.

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Evaluators Comments	Evaluator's Observation Marks Secured .... out of 50 Full Name of the Evaluator: Signature of the Evaluator: Date of Evaluation:

Case Study 8

Social Media Campaign Performance

Date of the Session: ....

Learning outcomes:

- Identify data type and scale.
- Compute mean, median, and mode.
- Analyze dispersion (variance, SD, IQR).
- Detect outliers and assess process stability.

8.1 Pre-Lab

1. What are discrete and continuous data?

A) Discrete data → Countable Ex: whole numbers. (7)

Continous data → Measurable values. Ex: Height 6.7

2. What are variance and standard deviation?

A) ⇒ Variance is, the avg of squared differences from the mean.

⇒ Square root of variance is std Dev.

3. How do outliers affect process stability?

A) i) Increase variation

ii) Disort avg

iii) Poor decision making.

8.2 Problem Statement

A company tracks the number of likes in its recent posts: 100, 120, 140, 160, 180, 200, 220, 250, 300, 5000 One post has gone viral, receiving significantly higher engagement.

1. Identify the data type and scale.

A) Data type = Discreat

scale : Ratio (0, Likes is a meaningful true zero)

2. Compute mean and median on likes of the post.

A) Mean = 667.0

Median = 190.0

3. Analyse the impact of the viral post on the average.

A) Mean with viral post = 667.0

Mean without viral post = 185.56

Impact : viral post inflated the mean by 481.44 likes

Median is robust → not affected.

4. Calculate dispersion measures (range, standard deviation, IQR).

A) range = 4900

Standard deviation = 1523.6055363519471

Q1 = 145.0 & Q3 = 242.5 IQR = 97.5

5. Identify outliers.

Lower bound = -1.25

upper bound = 388.75

6. Recommend which measure should be used for reporting performance.

A) Recommend measure for reporting : Median = 190.0

Reason : Mean is heavily skewed by viral post (5K likes),

Median reflects typical post performance more accurately.

Case Study 9

Income Distribution and Policy Planning

Date of the Session: ....

Learning outcomes:

- Identify and classify income data based on type (quantitative) and measurement scale (ratio scale), and understand its relevance in socio-economic analysis
- Compute and interpret key statistical measures including mean, median, and quartiles to summarize income distribution effectively.
- Analyze income inequality using dispersion measures such as range, interquartile range (IQR), and standard deviation, and interpret their implications in real-world contexts.

9.1 Pre-Lab

1. What is the difference between qualitative and quantitative data? Give examples related to income?

A) Qualitive means descriptive of words.

Ex:- Dhoni

Quantitative means numbers

Ex:- 07

2. Why is the median often preferred over the mean in income distribution analysis?

A) Median is preferred over mean because, Income data is usualy uneven (skewed).

=> Mean get Effected by Extream values

=> Median shows middle value.

3. What do you understand by income inequality? Which statistical tools help measure it?

A) Income InEquality means Income is not distributed equally among the people. The statistical tools are,
i) Gini Coefficient
ii) Lorenz Curve
iii) Percentile Ratios.

9.2 Problem Statement

A survey records monthly incomes (Rs.) of households: 8000, 10000, 12000, 15000, 18000, 20000, 25000, 30000, 100000 The government wants to classify income groups for welfare schemes. Do the following Tasks:

1. Identify the type and scale of data.

A) Data type = quantitative / Continous
Scale = Ratio. (Rs. 0 income is meaning full - true zero)

2. Compute mean, median, and quartiles.

A) Mean = 26444.444444444445
Median = 18000.0
Q_1 = 12000.0 ; Q_2 = 18000.0 ; Q_3 = 25000.0

3. Analyse income inequality using dispersion measures.

A) Range = 92000
IQR = 13000.0
Std Dev = 284.5494324772468
Interpretation: High std Dev = 285.55 → wide Income inequality present

4. Identify outliers.

A) Lower bound = -7500.0
Upper bound = 44500.0

5. Define "middle-income group" using quartiles.

A) Middle income group: 12000-25000 (b/w Q_1 & Q_3)

	House-hold	Income
2	H3	12,000
3	H4	15,000
4	H5	18000
5	H6	20,000
6	H7	25,000

Roll No :

6. Recommend which statistical measure should guide policy decisions.

A) Recommend measure = Median = 18000.0

Reasons : Mean is inflated by outliers household (Rs 1,00,000)

Median better reflects the typical households income for policy.

7. Discuss implications for economic planning.

A) Economic planning Implications

i) Welfare schemes → target households below Q1 (Rs 12000)

ii) Subsidies and tax relief for RS. 12000 RS 25000 bracket (middle class)

iii) High-Income outliers (RS. 1L) may reflect data anomaly con outliers wealth.

iv) use Gini Coefficient for a comprehensive inequality measurement.

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Evaluators Comments	**Evaluator's Observation** **Marks Secured .... out of 50** **Full Name of the Evaluator:** **Signature of the Evaluator:** **Date of Evaluation:**

Case Study 10

Exam Duration and Student Behavior

Date of the Session: ....

Learning outcomes:

- Identify data type and measurement scale.
- Compute mean, median, and standard deviation.
- Determine quartiles, IQR, and detect outliers.
- Interpret results to evaluate exam duration and consistency.
- Interpret results to evaluate exam duration and consistency.

10.1 Pre-Lab

1. What is the difference between qualitative and quantitative data? Give example

A) Qualitative data means → desciptive words Ex: good

Quantitative data means → numbers.

Ex: Thala → qualitative 7 → Quantitative.

2. What are quartiles and interquartile range (IQR)? Why are they important?

A) Quartiles are divided into 4 parts,

Q1 → 25%

Q2 → 50% (Median)

Q3 → 75%

IQR = Q3 − Q1

They are important because,

i) Helps understand data spread

ii) Detect outliers.

3. What is meant by a ratio scale of measurement? How is it different from interval scale?

A) Ratio Scale has true zero and supports "twice/half" Compairsions, inturval scale dose not.

10.2 Problem Statement

An instructor records time (minutes) taken by students to complete an exam: 30, 35, 40, 45, 50, 55, 60, 65, 70, 95. Perform the following Tasks:

1. Identify the data type and scale.

A) Data type: quantitative | Continous
Scale: Ratio (0 min = no time spent, true zero).

2. Compute mean, median, and standard deviation.

A) Mean = 54.5
Median = 52.5
Standard duration = 19.31371038 260613.

3. Determine quartiles and IQR.

A) Q1 = 41.25 IQR = 22.5
Q2 = 52.5
Q3 = 63.7

4. Identify outliers.

A) Lower bound = 7.5
upper bound = 97.5

5. Analyse whether the exam duration is appropriate.

A) Most of the students finish b/w Q1 = 41.25 & Q3 = 63.75 min. Mean = 54.5 min → a 60 min Exam is appropriate for most students.

Student taking an outlier is 95 min - may need Extra time (or) found it difficult

6. Recommend optimal exam timing.

A) The ReCommened duration is 75 (or 73 min. (Q_3 + 10 min buffer)

This accomidates 75% + students Comfortabely.

7. Evaluate consistency among students

A) CV = 35.25%

Moderate variability: mix of fast & slow students.

Suggestion: provide practice tests to reduce timing spread.

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Case Study 11

Warehouse Demand Forecasting

Date of the Session: ....

Learning outcomes:

- Identify data type (quantitative, discrete) and measurement scale (ratio).
- Detect outliers and analyse impact of demand spikes.
- Recommend forecasting methods and inventory strategies.

11.1 Pre-Lab

1. Effect of sudden demand spike on forecasting?

A) A sudden demand spike makes forecast unreliable and often overly optimistic.

2. What is demand forecasting? Name methods?

A) Demand fore casting means, Estimating the future demand using past data.

i) Qualitative ii) Quantitative.

3. Key factors in inventory planning? and Why handle irregular demand patterns?

A) Good inventory planning balances supply with demand especially when demand is unpredictable.

i) Avoid out of stock

ii) " over stock.

11.2 Problem Statement

Daily demand for a product: 50, 55, 60, 65, 70, 75, 80, 85, 90, 250. Do the following tasks:

1. Identify the data type and scale.

A) Data type : quantitative & Discreat

Scale : Ratio (0 units demanded is meaningful true zero)

2. Compute mean and median demand.

A)

Mean = 88.0

Median = 72.5

3. Analyse the effect of sudden demand spike.

A) Mean with spike = 88.0

Mean without spike = 70.0

spike inflated by mean is 18.0 units.

⇒ using mean for forcasting may lead to overstocking.

4. Calculate standard deviation and IQR.

A) std Dev = 58.36665714829543

Q_1 = 61.25 Q_2 = 72.5

Q_3 = 83.75 IQR = 22.5

5. Identify outliers.

A) Lower bound = 27.5

upper bound = 117.5

6. Recommend a suitable demand forecasting measure.

A) Recommended measure : MEDIAN 72.5 units

Reason : Mean is distorted by spike. Median gives realistic daily stock Estimate

For spiks : use moving Avg with outlier Exclusion.

7. Suggest inventory planning strategies.

A) Recommended safety stock = Median + 1.5 × SD = 160.0 units.

Inventory Strategies:

i) Mantian safty stocks to handel demand spikes.

ii) Use (JIT) for normal steady days

iii) Trigger alert when demand Exceeds upperfence (118 units).

iv) Consider Exponential smoothing (or ARIMA for time series forecasting.

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